

$\bar{D}_K$ is small. The opposite conclusions follow from “reverse Roy” selection where units with the smallest effects select into treatment first, which describes the take up of housing vouchers (Chyn forthcoming) and charter school applications (Walters forthcoming).

An easy way to gauge whether VWATT and a sample-weighted ATT is to scatter the weights from (13) against each group’s sample share. These two may be close if there is little variation in treatment timing, if the untreated group is very large, or if some timing groups are very large. Conversely, weighting matters less if the ATT_k ’s are similar, which one can evaluate by aggregating each group’s 2x2 DD estimates from the decomposition theorem.¹⁹ Finally, one could directly compare VWATT to point estimates of a particular parameter of interest. Several alternative estimators can deliver differently weighted averages of ATT’s (Abraham and Sun 2018, Callaway and Sant’Anna 2018, de Chaisemartin and D’Haultfœuille 2018b).

ii. Effects that vary over time but not across units

Time-varying treatment effects (even if they are identical across units) generate cross-group heterogeneity in VWATT by averaging time-varying effects over different post-treatment windows, but more importantly they mean that $\Delta ATT \neq 0$. Equations (11b) and (11c) show that common trends in counterfactual outcomes leaves one set of timing terms biased ($\hat{\beta}_{k\ell}^{2x2,\ell}$), while common trends between counterfactual and treated outcomes leaves the other set biased ($\hat{\beta}_{k\ell}^{2x2,k}$).

To illustrate this point, figure 3 plots a case where counterfactual outcomes are identical, but the treatment effect is a linear trend-break, $Y_{it}^1 = Y_{it}^0 + \phi \cdot (t - t_i^* + 1)$ (see Meer and West 2013). $\hat{\beta}_{k\ell}^{2x2,k}$ uses group ℓ as a control group during its pre-period and identifies the ATT during

¹⁹ Relatedly, de Chaisemartin and D’Haultfœuille (2018b) derive a statistic that gives the minimum variation in treatment effects that could lead to wrong-signed regression estimates when all treatment effects have the same sign.

the middle window in which treatment status varies: $\phi \frac{(t_\ell^* - t_k^* + 1)}{2} \cdot \hat{\beta}_{k\ell}^{2x2,\ell}$, however, is biased because the control group (k) experiences a trend in outcomes due to the treatment effect:²⁰

$$\beta_{k\ell}^{2x2,\ell} = \overbrace{ATT_\ell(POST(\ell))}^{\phi \frac{(T - (t_\ell^* - 1))}{2}} - \phi \frac{(T - (t_k^* - 1))}{2} = \phi \frac{(t_k^* - t_\ell^*)}{2} \leq 0 \quad (14)$$

This bias feeds through to $\beta_{k\ell}^{2x2}$ according to the size of $(1 - \mu_{k\ell})$:

$$\beta_{k\ell}^{2x2} = \phi \frac{[(2\mu_{k\ell} - 1)(t_\ell^* - t_k^*) + 1]}{2} \quad (15)$$

The entire two-group timing estimate can be wrong signed if $\mu_{k\ell}$ is small (putting more weight on the downward biased term). In figure 3, for example, both units are treated equally close to the ends of the panel, so $\mu_{k\ell} = 0.5$ and the estimated DD effect equals $\frac{\phi}{2}$, even though both units experience treatment effects as large as $\phi \cdot [T - (t_k^* - 1)]$. This type of bias affects the overall DD estimate according to the weights in equation (7). It is smaller when there are untreated units, more so when these units are large.

Note that this bias is specific to the specification in equation (2). More flexible event-study specifications may not suffer from this problem (although see Proposition 2 in Abraham and Sun

²⁰ The average of the effects for group k during any set of positive event-times is just ϕ times the average event-time. The $MID(k, \ell)$ period contains event-times 0 through $t_\ell^* - t_k^* - 1$ and the $POST(\ell)$ period contains event-times $t_\ell^* - (t_k^* - 1)$ through $T - (t_k^* - 1)$, so we have:

$$\begin{aligned} ATT_k(MID(k, \ell)) &= \phi \frac{(t_\ell^* - t_k^*)(t_\ell^* - t_k^* + 1)}{2(t_\ell^* - t_k^*)} = \phi \frac{t_\ell^* - t_k^* + 1}{2} \\ ATT_k(POST(\ell)) &= \phi(t_\ell^* - t_k^*) + \phi \frac{T - t_\ell^* + 2}{2} \end{aligned}$$

And the difference, which appears in the identifying assumption in (11c) equals:

$$ATT_k(POST(\ell)) - ATT_k(MID(k, \ell)) = \phi(t_\ell^* - t_k^*) + \phi \frac{T - t_\ell^* + 2}{2} - \phi \frac{t_\ell^* - t_k^* + 1}{2} = \frac{\phi}{2}(T - (t_k^* - 1))$$

Another way to see this, as noted in figure 3, is that average outcomes in the treatment group are always below average outcomes in the early group in the $POST(\ell)$ period and the difference equals the maximum size of the treatment effect in group k at the end of the $MID(k, \ell)$ period: $\phi \cdot (t_\ell^* - t_k^* + 1)$. Average outcomes for the late group are also below average outcomes in the early group in the $MID(k, \ell)$ period, but by the *average* amount of the treatment effect in group k during the $MID(k, \ell)$ period: $\phi \frac{(t_\ell^* - t_k^* + 1)}{2}$. Outcomes in group ℓ actually fall on average relative to group k , which makes the DD estimate negative even when all treatment effects are positive.

2018). Fadlon and Nielsen (2015) and Deshpande and Li (2017) use a novel estimator that matches treated units with controls that receive treatment a given amount of time later. This specification does not use already treated units as controls, and yields an average of $\hat{\beta}_{k\ell}^{2 \times 2, k}$ terms with a fixed post-period. Callaway and Sant'Anna (2018) discuss how to summarize heterogeneous treatment effects in this context and develop a reweighting estimator to do so. Summarizing time-varying effects using equation (2), however, yields estimates that are too small or even wrong-signed, and should not be used to judge the meaning or plausibility of effect sizes.²¹

B. What is the identifying assumption and how should we test it?

The preceding analysis maintained the assumption of *equal* counterfactual trends across groups, but (12) shows that (as long as treatment effects do not vary over time) identification of VWATT only requires VWCT to equal zero. The assumption itself is untestable because we cannot, for example, observe changes in Y^0 that occur after treatment or test for them using pre-treatment data.²² Assuming that differential counterfactual trends, ΔY_k^0 , are linear throughout the panel leads to a convenient approximation to VWCT:²³

$$\sum_{k \neq U} \Delta Y_k^0 \left[\sigma_{kU} + \sum_{j=1}^{k-1} \sigma_{jk} (1 - 2\mu_{jk}) + \sum_{j=k+1}^K \sigma_{kj} (2\mu_{kj} - 1) \right] - \Delta Y_U^0 \sum_{k \neq U} \sigma_{kU} = 0 \quad (16)$$

²¹ Borusyak and Jaravel (2017) show that that common, linear trends, in the post- *and* pre- periods cannot be estimated in this design. The decomposition theorem shows why: timing groups act as controls for each other, so permanent common trends difference out. This is not a meaningful limitation for treatment effect estimation, though, because “effects” must occur after treatment. Job displacement provides a clear example (Jacobson, LaLonde, and Sullivan 1993, Krolikowski 2017). Comparisons based on displacement timing cannot identify whether all displaced workers have a permanently different earnings trajectory than never displaced workers (the unidentified linear component), but they can identify changes in the time-path of earnings around the displacement event (the treatment effect).

²² One can use time-varying confounders as outcomes (Freyaldenhoven, Hansen, and Shapiro 2018, Pei, Pischke, and Schwandt 2017), but this does not test for balance in levels, nor can it be used for sparsely measured confounders.

²³ The path of counterfactual outcomes can affect each 2x2 DD’s bias term in different ways. Linearly trending unobservables, for example, lead to larger bias in 2x2 DD’s that use more periods. I show in the appendix that in the linear case differences in the magnitude of the bias cancel out across each group’s “treatment” and “control” terms, and equation (16) holds.

Equation (16) generalizes the definition of common trends and collapses to the typical pairwise common trends assumption for any two-group estimator. It is immediately clear that *identical* trends across all groups satisfies (16), but this is not a necessary condition for it to hold since the bias induced by a given group's trend depends on the weights in brackets.²⁴

Fortunately, the weights have an intuitive interpretation. The weight on each group's counterfactual trend equals the difference between the total weight it gets when it acts as a treatment group— w_k^T from equation (13)—minus the total weight it gets when it acts as a control group— w_k^C . $VWCT$ can therefore be written as:

$$\sum_k \Delta Y_k^0 [w_k^T - w_k^C] = 0 \quad (17)$$

Figure 4 plots $w_k^T - w_k^C$ as a function of $\bar{D}$ (assuming equal group sizes). Units treated in the middle of the panel have high treatment variance and get a lot of weight when they act as the treatment group, while units treated toward the ends of the panel get relatively more weight when they act as controls. As t^* moves closer to 1 or T , $w_k^T - w_k^C$ becomes negative, which shows that some timing groups effectively act as controls. This helps define “the” control group in timing-only designs (the dashed line in figure 6): all groups are controls in *some* terms, but the earliest and/or latest units necessarily get more weight as controls than treatments. The weights also map differential trends to bias.²⁵ A positive trend in group k induces positive bias when $w_k^T - w_k^C > 0$, negative bias when $w_k^T - w_k^C < 0$ (that is, if k is an effective control group), and no bias when $w_k^T - w_k^C = 0$. The size of the bias from a given trend is larger for groups with more weight.

²⁴ Callaway and Sant'Anna (2018) provide alternative definitions of common trends and tests not based on linearity.

²⁵ Applications typically discuss bias in general terms, arguing that unobservables must be “uncorrelated” with timing, but have not been able to specify *how* counterfactual trends would bias a two-way fixed effects estimate. For example, Almond, Hoynes, and Schanzenbach (2011, p 389-190) argue: “Counties with strong support for the low-income population (such as northern, urban counties with large populations of poor) may adopt FSP earlier in the period. This systematic variation in food stamp adoption could lead to spurious estimates of the program impact if those same county characteristics are associated with differential trends in the outcome variables.”